

Getting structured output: JSON format responses

In some cases, you may want OpenAI to provide a structured response, especially if you're building applications that require data to be formatted in a specific way. OpenAI can generate responses in formats like JSON, which is useful for assessments, forms, or APIs.

When requesting answers in JSON format, it's essential to specify both the structure and the content requirements clearly within the prompt. This ensures the AI understands the desired format and the type of data it needs to include.

Here's an example prompt for the JSON format.

- **Prompt:** "Generate three multiple-choice AI questions for an assessment, along with four answer options for each question, and provide the correct answer in JSON format. The JSON format should follow this structure:

```
{
  "questions": [
    {
      "question": "Question text",
      "options": ["Option A", "Option B", "Option C", "Option D"],
      "correct_answer": "Correct option"
    }
  ]
}
```

Provide the data in this format for each question."

The example response is:

```
{
  "questions": [
    {
      "question": "What is a neural network?",
      "options": [
        "A system designed to mimic the human brain",
        "A type of network used for data transmission",
        "A social media platform",
        "An algorithm for decision trees"
      ],
      "correct_answer": "A system designed to mimic the human brain"
    },
    {
      "question": "Which of the following is a type of machine learning?",
      "options": [
        "Supervised learning",
        "Quantum computing",
        "Database management",
        "Firewall configuration"
      ]
    }
  ]
}
```

```
],
  "correct_answer": "Supervised learning"
},
{
  "question": "Which company developed GPT-3?",
  "options": [
    "Google",
    "Microsoft",
    "OpenAI",
    "Amazon"
  ],
  "correct_answer": "OpenAI"
}
]
```

Why does this work? By specifying the exact structure of the JSON in the prompt, you ensure that OpenAI generates a response that can be easily parsed by a system or used in automated workflows. This approach is particularly helpful when working with AI in application development or form-based assessments.

The best practices are:

- **Specify the format:** Be explicit about the data structure (in this case, JSON) and the keys you want in the response.
- **Test the output:** Ensure the AI's response aligns with the format by tweaking the prompt if necessary.
- **Use with applications:** This is ideal for integrating AI-generated data directly into APIs or databases that require structured input.

Getting the right answers from OpenAI's models comes down to how you frame the questions. With efficient prompt engineering, you can guide the AI to provide clear, relevant, and targeted responses. By preparing thoughtful inputs, planning the desired output, and experimenting with various prompts, you can unlock the full potential of OpenAI's language models. Remember, the more specific and structured your prompts, the better the answers you'll receive. **END** 🐧

 **By: Somanath Balakrishnan**

The author has 20+ years of experience in software development and is currently working as a software architect at Digital Core Technologies, Kochi.

Disclaimer: This article heavily used OpenAI services to generate the content and improve quality because OpenAI knows the context of prompt engineering better than us to get the right answers. LOL!